ADVANCE PYTHON

CONDITIONAL STATEMENTS

In [1]:
```
if True:    #indentiation is always 4 space
    print("Data Science")
```

```
Data Science
```

In [3]:
```
if False:
    print("Data Science")
print("Byy for now")
```

```
Byy for now
```

In [4]:
```
if True:
    print("Data Science")
print("Byy for now")
```

```
Data Science
Byy for now
```

In [5]:
```
if True:
    print("Data Science")
else:
    print("Byy for now")
```

```
Data Science
```

Check the number is even or odd Number

In [19]:
```
x = 7
r = x % 2

if r == 0:
    print("Even Number")
```

In [13]:
```
x = 7
r = x % 2

if r == 0:
    print("Even Number")
else:
    print("Odd Number")
```

```
Odd Number
```

In [3]:
```
x = 8
r = x % 2
```

```
if r == 0:
    print("Even Number")

else:
    print("Odd Number")
```

```
Even Number
```

pls check number is even or odd # pls check number is greater or smaller

NESTED IF - ELSE

In [17]:

```
x = 3
r = x % 2

if r == 0:
    print("Even Number")
    if x > 5:
        print("Greater Number")

else:
    print("Odd Number")
```

```
Odd Number
```

In [22]:

```
x = 4
r = x % 2

if r == 0:
    print("Even Number")
    if x > 5:
        print("Greater Number")
    else:
        print("Smaller Number")

else:
    print("Odd Number")
```

```
Even Number
Smaller Number
```

In [23]:

```
x = 8
r = x % 2

if r == 0:
    print("Even Number")
    if x > 5:
        print("Greater Number")
    else:
        print("Smaller Number")

else:
    print("Odd Number")
```

```
Even Number
Greater Number
```

IF - ELIF - ELSE

In [28]:
```
x = 2

if x == 1:
    print("one")
elif x == 2:
    print("Two")
elif x == 3:
    print("Three")
elif x == 4:
    print("Four")
```

```
Two
```

In [31]:
```
x = 25

if x == 1:
    print("one")
elif x == 2:
    print("Two")
elif x == 3:
    print("Three")
elif x == 4:
    print("Four")

else:
    print("Number not found")
```

```
Number not found
```

LOOPS

In programming world some time we keep on repeating, may be you want to repeat 5 statements so 1. WHILE LOOP 2. FOR LOOP

In [2]:
```
i = 1         #initializing

while i <= 5:           #condition
    print("Data Science")
    i = i+1         #increment
```

```
Data Science
Data Science
Data Science
Data Science
Data Science
```

In [5]:
```
i = 5         #initializing

while i >= 1:           #condition
```

```
    print("Data Science", ':', i)
    i = i - 1        #increment
```

```
Data Science : 5
Data Science : 4
Data Science : 3
Data Science : 2
Data Science : 1
```

In [6]:
```
i = 20       #initializing

while i >= 1:           #condition
    print("Data Science", ':', i)
    i = i - 1        #increment
```

```
Data Science : 20
Data Science : 19
Data Science : 18
Data Science : 17
Data Science : 16
Data Science : 15
Data Science : 14
Data Science : 13
Data Science : 12
Data Science : 11
Data Science : 10
Data Science : 9
Data Science : 8
Data Science : 7
Data Science : 6
Data Science : 5
Data Science : 4
Data Science : 3
Data Science : 2
Data Science : 1
```

In [7]:
```
#Nested While loop

i = 1
while i <=5:
    print("Data Science")   #when we mention end then new linw will not create
    j = 1
    while j <=4:
        print("IT Technology")
        j = j + 1

    i = i + 1
    print()
```

```
Data Science
IT Technology
IT Technology
IT Technology
IT Technology

Data Science
IT Technology
IT Technology
IT Technology
IT Technology

Data Science
IT Technology
IT Technology
IT Technology
IT Technology

Data Science
IT Technology
IT Technology
IT Technology
IT Technology

Data Science
IT Technology
IT Technology
IT Technology
IT Technology
```

In [8]:

```
#Nested While loop

i = 1
while i <=5:
    print("Data Science", end = ' = ')    #when we mention end then new linw will no
    j = 1
    while j <=4:
        print("IT Technology", end = ' = ')
        j = j + 1

    i = i + 1
    print()
```

```
Data Science = IT Technology = IT Technology = IT Technology = IT Technology =
Data Science = IT Technology = IT Technology = IT Technology = IT Technology =
Data Science = IT Technology = IT Technology = IT Technology = IT Technology =
Data Science = IT Technology = IT Technology = IT Technology = IT Technology =
Data Science = IT Technology = IT Technology = IT Technology = IT Technology =
```

In [13]:

```
# let use while loop using some numbers

i = 1
while i <= 2:
    j = 0
    while j <=2:
```

```
        print(i*j, end = " ")
        j +=1
    print()
    i += 1
```

```
0 1 2
0 2 4
```

In [14]:

```
i = 1
while i <= 4:
    j = 0
    while j <= 3:
        print(i*j, end = " ")
        j +=1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

In [15]:

```
name = "Alfiya"
for i in name:
    print(i)
```

```
A
l
f
i
y
a
```

In [16]:

```
name = [1,2,3]
for i in name:
    print(i)
```

```
1
2
3
```

In [17]:

```
range(5)
```

Out[17]:

```
range(0, 5)
```

In [19]:

```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

In [20]:

```
for i in range(1,10,3):
    print(i)
```

```
1
4
7
```

In [23]:
```python
#print the table  of 5

for i in range(1, 51):

    if i % 5 == 0:
        print(i)
```

```
5
10
15
20
25
30
35
40
45
50
```

In [1]:
```python
import keyword
keyword.kwlist
```

```
Out[1]: ['False',
 'None',
 'True',
 'and',
 'as',
 'assert',
 'async',
 'await',
 'break',
 'class',
 'continue',
 'def',
 'del',
 'elif',
 'else',
 'except',
 'finally',
 'for',
 'from',
 'global',
 'if',
 'import',
 'in',
 'is',
 'lambda',
 'nonlocal',
 'not',
 'or',
 'pass',
 'raise',
 'return',
 'try',
 'while',
 'with',
 'yield']
```

In [2]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

In [3]:
```
for i in range(1,11):
    if i == 6:
        break
    print(i)
```

```
1
2
3
4
5
```

In [4]:

```
for i in range(1,11):
    if i == 6:
        continue
    print(i)
```

```
1
2
3
4
5
7
8
9
10
```

In [5]:

```
for i in range(1,11):
```

```
  Cell In[5], line 1
    for i in range(1,11):
                         ^
_IncompleteInputError: incomplete input
```

In [6]:

```
for i in range(1,11):
    pass
```

In [7]:

```
#write the code ask choclate from vendor machne write the basic code
x = int(input("How many choclate you want?"))

i = 1
while i <= x:
    print("Choclets")
    i+= 1
```

```
Choclets
Choclets
Choclets
Choclets
Choclets
```

In [10]:

```
available_choclate = 5    #Vendor machine only has 10 candies
x = int(input("How many choclate you want?"))

i = 1
while i <= x:
    if i >available_choclate:
        break
    print("Choclets")
    i+= 1
print("Out of stock")
```

```
Choclets
Choclets
Choclets
Choclets
Choclets
Out of stock
```

Print number from 1 to 50 but dont print number which is divisible by 3 or 5

In [12]:
```python
for  i in range(1,51):
    if i %3 == 0:
        print(i)
print("End")
```

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
End
```

In [14]:
```python
for  i in range(1,51):
    if i %3 == 0:
        continue
    print(i)
print("End")
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
End
```

In [15]:

```
for  i in range(1,51):
    if i %3 == 0 or i %5 == 0:
        continue
    print(i)
print("End")
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
End
```

In [16]:

```python
# when you apply and you wont get the value which is divisible by both 3 & 5 (15)

for i in range(1,50):
    if i%3 == 0 or i%5 == 0:
        continue
    print(i)
print('end')
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end
```

In [17]:

```
# i dont want to print the values which are even numbers that means print only odd
for i in range(1,51):

    if (i%2 == 0):
        #print('even')
        continue
    else:
        print(i)
print('bye')
```

```
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
bye
```

FOR ELSE

In [28]:

```python
nums = [12,15,18,21,26]

for num in nums:
    if num % 5 == 0:
        print(num)
```

```
15
```

In [29]:

```python
nums = [12,15,18,21,26,20]

for num in nums:
    if num % 5 == 0:
        print(num)
```

```
15
20
```

In [30]:

```python
nums = [12,18,21,26]

for num in nums:
    if num % 5 == 0:
        print(num)
```

In [33]:

```python
nums = [12,18,21,26]

for num in nums:
```

```python
    if num % 5 == 0:
        print(num)
    else:
        print("number not found")
```

```
number not found
number not found
number not found
number not found
```

In [34]:
```python
nums = [12,18,21,26]

for num in nums:
    if num % 5 == 0:
        print(num)
else:
    print("number not found")
```

```
number not found
```

In [47]:
```python
num = 14

for i in range(2,num):
    if num % i == 0:
        print('Not prime Number')
        break
else:
    print('Prime Number')
```

```
Not prime Number
```

In [48]:
```python
num = 13

for i in range(2,num):
    if num % i == 0:
        print('Not prime Number')
        break
else:
    print('Prime Number')
```

```
Prime Number
```

PRINTING PATTERN IN PYTHON

In [20]:
```python
print("# # # #")
print("# # # #")
print("# # # #")
print("# # # #")
print("# # # #")
```

```
# # # #
# # # #
# # # #
# # # #
# # # #
```

In [21]:
```
for i in range(1,5):
    i=i+1
    print('# # # # ')
```

```
# # # #
# # # #
# # # #
# # # #
```

In [22]:
```
for i in range(1,5):
    if i<=5:
        print('* * * *')
```

```
* * * *
* * * *
* * * *
* * * *
```

In [23]:
```
for j in range(4):
    print('@')
```

```
@
@
@
@
```

In [24]:
```
for j in range(4):
    print('* * * *')
```

```
* * * *
* * * *
* * * *
* * * *
```

In [25]:
```
for j in range(4):
    print('#', end = " ")

print()

for j in range(4):
    print('#', end = " ")

print()

for j in range(4):
    print('#', end = " ")

print()

for j in range(4):
    print('#', end = " ")
```

```
# # # #
# # # #
# # # #
# # # #
```

In [26]:
```
for j in range(4):
    print('*', end=" ")

for j in range(4):
    print('*', end=" ")
```

```
* * * * * * * *
```

In [27]:
```
for j in range(4):
    print('#', end="  ")

print()

for j in range(4):
    print('#', end="  ")

print()

for j in range(4):
    print('#', end="  ")

print()

for j in range(4):
    print('#', end="  ")
```

```
#  #  #  #
#  #  #  #
#  #  #  #
#  #  #  #
```

In [35]:
```
for i in range(4):
    for j in range(i+1):
        print("#", end = "  ")
    print()
```

```
#
#  #
#  #  #
#  #  #  #
```

In [41]:
```
for i in range(1,5):
    print(" # " * i)
```

```
 #
 #  #
 #  #  #
 #  #  #  #
```

In [42]:
```
list(range(5))
```

Out[42]:
```
[0, 1, 2, 3, 4]
```

In [43]:
```
for i in range(4):
    for j in range(i):
```

```
        print('#', end="  ")
    print()
```

```
#
#  #
#  #  #
```

In [44]:
```
for i in range(4):
    for j in range(i+1):
        print('#', end="  ")
    print()
```

```
#
#  #
#  #  #
#  #  #  #
```

In [45]:
```
for i in range(4):
    for j in range(4-i):
        print('#', end="  ")
    print()
```

```
#  #  #  #
#  #  #
#  #
#
```

In [46]:
```
for i in range(1,6):
    print("# "*(6-i))
```

```
# # # # #
# # # #
# # #
# #
#
```

1.Right Angle Triangle pattern

In [2]:
```
for i in range(1,6):
    print(' * ' *i)
```

```
 *
 *  *
 *  *  *
 *  *  *  *
 *  *  *  *  *
```

2.Inverted Right Angle Triangle

In [5]:
```
for i in range(5,0,-1):
    print(' * '*i)
```

```
*  *  *  *  *
*  *  *  *
*  *  *
*  *
*
```

3.Pyramid Pattern

In [7]:
```
for i in range(1,6):
    print(''*(5-i)+' * '*(2*i-1))
```

```
*
*  *  *
*  *  *  *  *
*  *  *  *  *  *  *
*  *  *  *  *  *  *  *  *
```

4.Inverted Pyramid

In [8]:
```
for i in range(5,0,-1):
    print(''*(5-i)+' * '*(2*i-1))
```

```
*  *  *  *  *  *  *  *  *
*  *  *  *  *  *  *
*  *  *  *  *
*  *  *
*
```

5.Diamond pattern

In [9]:
```
for i in range(1,6):
    print(''*(5-i)+' * '*(2*i-1))
for i in range(4,0,-1):
    print(''*(5-i)+' * '*(2*i-1))
```

```
*
*  *  *
*  *  *  *  *
*  *  *  *  *  *  *
*  *  *  *  *  *  *  *  *
*  *  *  *  *  *  *
*  *  *  *  *
*  *  *
*
```

6.Hallow Square

In [12]:
```
for i in range(5):
    for j in range(5):
```

```
        if i==0 or i==4 or j==0 or j==4:
            print('*',end='')
        else:
            print(' ',end='')
    print()
```

```
*****
*   *
*   *
*   *
*****
```

7.Full Square Pattern

In [13]:
```
for i in range(5):
    print(' * '*5)
```

```
 *  *  *  *  * 
 *  *  *  *  * 
 *  *  *  *  * 
 *  *  *  *  * 
 *  *  *  *  * 
```

8.Right Angle Triangle(Number Pattern)

In [14]:
```
for i in range(1,6):
    print(' '.join(str(x) for x in range(1,i+1)))
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

9. Inverted Right Angle Triangle

In [15]:
```
for i in range(5,0,-1):
    print(' '.join (str(x) for x in range(1,i+1)))
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

10.Floyd's Triangle

In [17]:
```
num=1
for i in range(1,6):
    for j in range(1,i+1):
```

```
        print(num,end=' ')
        num+=1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

11.Hallow Right Angle Triangle

In [18]:
```
for i in range(1,6):
    for j in range(1,i+1):
        if j==1 or j==i or i==5:
            print('*',end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
*
* *
*   *
*     *
* * * * *
```

12.Hallow Pyramid Pattern

In [19]:
```
for i in range(1,6):
    for j in range(5-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j==0 or j==2*i-2 or i==5:
            print('*',end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
        *
      *   *
    *       *
  *           *
* * * * * * * * *
```

13.Hallow Diamond Pattern

In [20]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
```

```
        if j==0 or j==2*i-2:
            print('*',end=' ')
        else:
            print(' ',end=' ')
    print()

for i in range(n-1,0,-1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j == 0 or j == 2 * i - 2:
            print('*',end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
        *
      *   *
    *       *
  *           *
*               *
  *           *
    *       *
      *   *
        *
```

14.Hallow Diamond Number Pattern

In [21]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j==0 or j==2*i-2:
            print(i ,end=' ')
        else:
            print(' ',end=' ')
    print()

for i in range(n-1,0,-1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        if j == 0 or j == 2 * i - 2:
            print(i ,end=' ')
        else:
              print(' ',end=' ')
    print()
```

```
        1
      2   2
    3       3
  4           4
5               5
  4           4
    3       3
      2   2
        1
```

15.Butterfly Pattern

In [28]:
```
n=5
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2* (n-i)):
        print(' ', end = ' ')
    for  j in range(1, i + 1):
        print(j, end = ' ')
    print()

for i in range(n, 0, -1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2* (n-i)):
        print( ' ', end = ' ')
    for j in range(1, i + 1):
        print(j, end = ' ')
    print()
```

```
1                 1
1 2             1 2
1 2 3         1 2 3
1 2 3 4     1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4     1 2 3 4
1 2 3         1 2 3
1 2             1 2
1                 1
```

In [30]:
```
n=5
for i in range(1,n+1):
    for j in range(i):
        print('*', end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(i):
        print('*', end=' ')
    print()

for i in range(n,0,-1):
    for j in range(i):
```

```
        print('*',end=' ')
    for j in range(2*(n-i)):
        print(' ', end=' ')
    for j in range(i):
        print('*', end=' ')
    print()
```

```
*                 *
* *             * *
* * *         * * *
* * * *     * * * *
* * * * * * * * * *
* * * * * * * * * *
* * * *     * * * *
* * *         * * *
* *             * *
*                 *
```

16.Hallow Number Pyramid

In [31]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(1,2*i):
        if j==1 or j==2*i-1 or i==n:
            print(i,end=' ')
        else:
            print(' ',end=' ')
    print()
```

```
        1
      2   2
    3       3
  4           4
5 5 5 5 5 5 5 5 5
```

17.Full Star pyramid

In [32]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        print('*',end=' ')
    print()
```

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
```

18.Inverted Full Star Pyramid

In [33]:
```
n=5
for i in range(n,0,-1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(2*i-1):
        print('*',end=' ')
    print()
```

```
* * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        *
```

19.Left Aligned Pyramid Pattern

In [34]:
```
n=5
for i in range(1,n+1):
    for j in range(i):
        print('*',end=' ')
    print()
```

```
*
* *
* * *
* * * *
* * * * *
```

In [35]:
```
n=5
for i in range(1,n+1):
    for j in range(1,i+1):
        print(j,end=' ')
    print()
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

20.Right Aligned Pyramid Pattern

In [36]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(1,i+1):
```

```
        print(j,end=' ')
    print()
```

```
        1
      1 2
    1 2 3
  1 2 3 4
1 2 3 4 5
```

In [37]:
```
n=5
for i in range(1,n+1):
    for j in range(n-i):
        print(' ',end=' ')
    for j in range(i):
        print('*',end=' ')
    print()
```

```
        *
      * *
    * * *
  * * * *
* * * * *
```